

Credit Risk Assessment of a Healthcare Loan Portfolio (Portfolio 2)

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1 Introduction

Financial institutions are required to hold sufficient capital to protect against unexpected losses, and tail risk is heavily influenced by joint defaults. This report analyses the one-year credit risk of Portfolio 2, which consists of 60 healthcare obligors with a total exposure at default (EAD) of 1,454,724. Since a single-industry portfolio can suffer from a sector-wide shock hitting many borrowers simultaneously, measuring the dependence between these borrowers is central to the modelling. The approach combines rating-based probabilities of default (PD) with a one-factor Gaussian copula and Monte Carlo simulation, and benchmarks the results against the analytical Vasicek formula and the Basel IRB capital requirement. The expected loss (EL) is relatively low at roughly 9,600, representing 0.66% of EAD, but the 99.9% Value-at-Risk (VaR) reaches approximately 123,000. Dependence therefore roughly doubles the tail loss compared with the assumption of independent defaults (61,118).

2 Data and Descriptive Analysis

The portfolio composition was merged with the company information and the ratings on `CompanyID`, with each merge validated as one-to-one. All 60 obligors were matched, and there were no duplicates or missing ratings. The stock return data contain 1,274 daily observations (2018-01-01 to 2022-12-30) for the 60 firms, with 3.0% of values missing. The healthcare index has no missing values but contains one large outlier (+22.1% on 2022-03-23).

Rating	Obligors	Exposure	Share
AAA	2	56,368	3.9%
AA	2	45,182	3.1%
A	12	362,463	24.9%
BBB	18	413,164	28.4%
BB	17	387,900	26.7%
B	7	134,971	9.3%
CCC	2	54,675	3.8%
Total	60	1,454,724	100%

Table 1: Rating distribution of Portfolio 2.

Statistic	Value
Mean / max exposure	24,245 / 60,491
Top-10 share	31.3%
HHI	0.021
Effective N ($1/\text{HHI}$)	48.7
Gini coefficient	0.26

Table 2: Exposure concentration.

The portfolio's credit quality is divided (Table 1, Figure 1): 57% of the obligors are investment grade by count, but the 26 sub-investment-grade borrowers make up nearly 40% of the total exposure. The CCC-rated names represent only 3.8% of the exposure but carry the highest probability of default, leading to a disproportionate share of the expected loss; firm 90 alone accounts for 8% of the portfolio's expected loss. The Herfindahl–Hirschman Index of 0.021 (Table 2) corresponds to an effective number

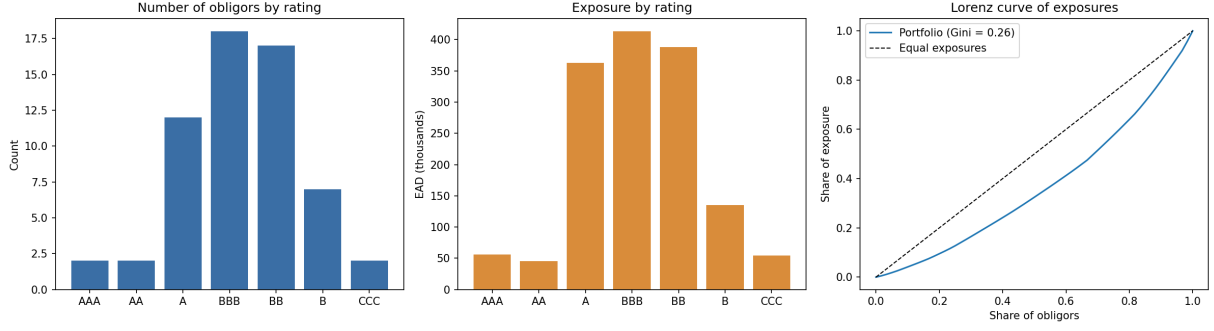


Figure 1: Obligor by rating, exposure by rating, and Lorenz curve of exposures.

of roughly 49 equal-sized loans compared to the actual 60, showing mild size dispersion with no single-name dominance, as the largest loan is only 4.2% of the total exposure. Missing returns were handled by pairwise deletion, which retains on average 94% of the observations for each pair of firms, whereas listwise deletion would keep only 205 of the 1,274 days because the gaps fall on different days for different firms. Although there were no missing ratings in this dataset, the code applies a fallback rule that assigns the worst observed rating, to remain conservative.

3 Model Setup and Assumptions

Defaults are modelled with a one-factor Gaussian copula (Vasicek, 2002). Each obligor i has a latent asset value

$$X_i = \sqrt{\rho_i} Z + \sqrt{1 - \rho_i} \varepsilon_i, \quad Z, \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, 1), \quad (1)$$

where Z is the systematic healthcare-sector factor and ε_i is the firm-specific shock. Obligor i defaults if its asset value falls below the threshold implied by its PD, and the portfolio loss is

$$D_i = \mathbf{1}\{X_i < \Phi^{-1}(PD_i)\}, \quad L = \sum_{i=1}^{60} EAD_i \cdot LGD \cdot D_i. \quad (2)$$

Conditional on the systematic factor $Z = z$, defaults are independent with probability

$$PD_i(z) = \Phi\left(\frac{\Phi^{-1}(PD_i) - \sqrt{\rho_i} z}{\sqrt{1 - \rho_i}}\right). \quad (3)$$

Parameter	Assumption	Source / reason
Horizon	1 year	1-year transition matrix
EAD	Given exposure	<code>portfolio2_composition.csv</code>
LGD	45% (fixed)	Basel F-IRB, senior unsecured
PD	Default column of transition matrix	AAA/AA floored at 0.03% (Basel floor)
ρ (main)	Basel IRB formula, 0.130–0.238	Section 3.2
Scenarios	200,000	seed 42; stability check in Section 4

Table 3: Model assumptions.

3.1 Probability of default

Rating	AAA	AA	A	BBB	BB	B	CCC
1-year PD	0.03%	0.03%	0.50%	1.20%	2.00%	3.00%	5.00%
3-year cum. PD	0.04%	0.18%	1.60%	3.54%	5.82%	9.11%	17.02%
5-year cum. PD	0.17%	0.57%	2.80%	5.83%	9.46%	14.72%	26.37%

Table 4: Probabilities of default. Multi-year values are taken from P^t , assuming a time-homogeneous Markov chain.

3.2 Dependence structure: three estimation approaches

Approach	Result
A. Index factor: $\rho_i = \text{corr}(r_i, r_{\text{index}})^2$	mean $\rho = 0.0009$; 4/60 significant at 5%
B. Pairwise: mean off-diagonal correlation; eigenvalues	-0.001 ; $\lambda_{\max} = 1.457 < \text{MP bound } 1.481$
C. Basel IRB (Basel Committee on Banking Supervision, 2005): $\rho = 0.12w + 0.24(1 - w)$, $w = (1 - e^{-50PD})/(1 - e^{-50})$	0.130–0.238; EAD-weighted 0.185

Table 5: Correlation estimates. MP = Marchenko–Pastur upper bound for the eigenvalues of a pure-noise correlation matrix.

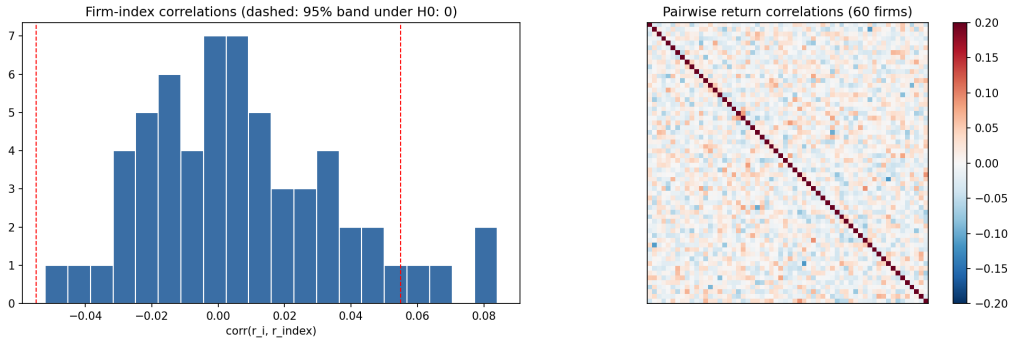


Figure 2: Left: firm–index correlations with the 95% band under $H_0: \rho = 0$. Right: pairwise correlation matrix.

The one-factor Gaussian copula is the industry standard for this type of analysis and underlies the Basel IRB formula (Gordy, 2003). It has few parameters, reproduces each obligor’s marginal default probability exactly, and drives dependence through a single interpretable sector factor. Under this framework a firm defaults when its asset value drops below a threshold matching its probability of default, so that the simulated default rate of each obligor equals its target PD. The empirical point estimates of the correlation are close to zero (Table 5, Figure 2): only 4 of 60 firms are significant at the 5% level, about what chance alone would produce, and the largest eigenvalue lies below the Marchenko–Pastur bound, so the correlation matrix is virtually indistinguishable from pure noise. Assuming zero correlation, however, would cut the capital requirement in half and heavily understate risk, since real-world evidence shows that sector-wide recessions cause clustered defaults. Relying on the Basel supervisory calibration for correlation is therefore the more prudent choice. A 3-basis-point floor is applied to the default probabilities to avoid the undefined threshold $\Phi^{-1}(0) = -\infty$ and to align with Basel standards.

4 Simulation and Risk Metrics

For each of 200,000 scenarios, one systematic factor and 60 idiosyncratic shocks are drawn and the portfolio loss is computed. VaR_q is the empirical q -quantile of the simulated losses, Expected Shortfall is $\text{ES}_q = \mathbb{E}[L \mid L \geq \text{VaR}_q]$, and economic capital is $\text{VaR}_q - \text{EL}$. Ten independent runs of 50,000 scenarios give a mean 99.9% VaR of 121,611 with a standard deviation of 3,292 (2.7%), so the simulation error with 200,000 scenarios is small.

Metric	Independent ($\rho = 0$)	Empirical ρ	Basel ρ (main)
Expected loss	9,625	9,624	9,635
Std. dev. (UL)	10,926	10,929	15,745
VaR 95%	30,884	30,868	40,489
ES 95%	39,347	39,349	59,783
VaR 99%	44,330	44,428	71,136
ES 99%	51,891	51,957	92,908
VaR 99.9%	61,118	61,468	123,170
ES 99.9%	68,365	68,379	147,745
Economic capital 99.9%	51,493	51,844	113,535
P(no default)	39.1%	39.2%	53.0%
Max. defaults	8	8	27

Table 6: Monte Carlo results, 200,000 scenarios.

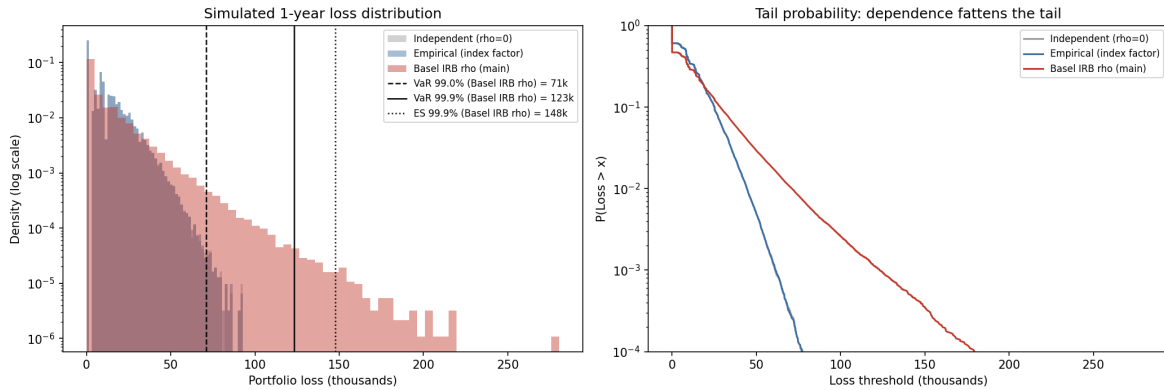


Figure 3: Simulated one-year loss distribution (log density, left) and exceedance probability $P(L > x)$ (right).

Expected loss is identical across the three models (Table 6) because expectation is linear and $\mathbb{E}[D_i] = PD_i$ regardless of ρ ; correlation changes only the shape of the loss distribution, not its mean. The variance, by contrast, contains covariance terms that grow with the correlation, raising the unexpected loss from roughly 10,900 to 15,700. The probability of zero defaults rises from 39% to 53% under positive correlation because good years become cleaner and bad years worse, shifting probability mass towards both ends of the distribution. The resulting fatter tail (Figure 3) raises the maximum number of simulated defaults from 8 to 27 and the 99.9% VaR from 61,000 to 123,000. Expected Shortfall always exceeds VaR because it averages the losses beyond the VaR threshold, which makes it a coherent risk measure that is more informative about the severity of tail events.

5 Extensions

Analytical VaR. In the asymptotic single risk factor (ASRF) limit, the portfolio loss converges to its conditional expectation given Z , so evaluating Equation (3) at the $(1 - q)$ -quantile of the factor gives

$$\text{VaR}_q^{\text{ASRF}} = \sum_i \text{EAD}_i \cdot \text{LGD} \cdot \Phi \left(\frac{\Phi^{-1}(\text{PD}_i) + \sqrt{\rho_i} \Phi^{-1}(q)}{\sqrt{1 - \rho_i}} \right), \quad (4)$$

which equals 98,850 at $q = 99.9\%$, compared with 123,170 from the simulation.

Basel IRB capital. With maturity $M = 2.5$, the IRB formula (Basel Committee on Banking Supervision, 2005) gives a capital requirement of $K = 109,924$ (7.56% of EAD), corresponding to risk-weighted assets of 1,374,049. This compares with a simulated economic capital of 113,535.

ρ	0.00	0.02	0.05	0.10	0.15	0.20	0.30	0.50
MC VaR 99.9%	60.2	65.9	73.3	92.4	113.4	136.5	193.4	312.0
MC ES 99.9%	67.9	73.5	83.3	108.2	136.3	162.9	232.8	380.1
Vasicek VaR 99.9%	9.6	24.8	39.4	62.9	87.0	112.3	167.0	294.3

Table 7: Correlation sensitivity in thousands, homogeneous ρ , 100,000 scenarios. EL stays at about 9.6 thousand throughout.

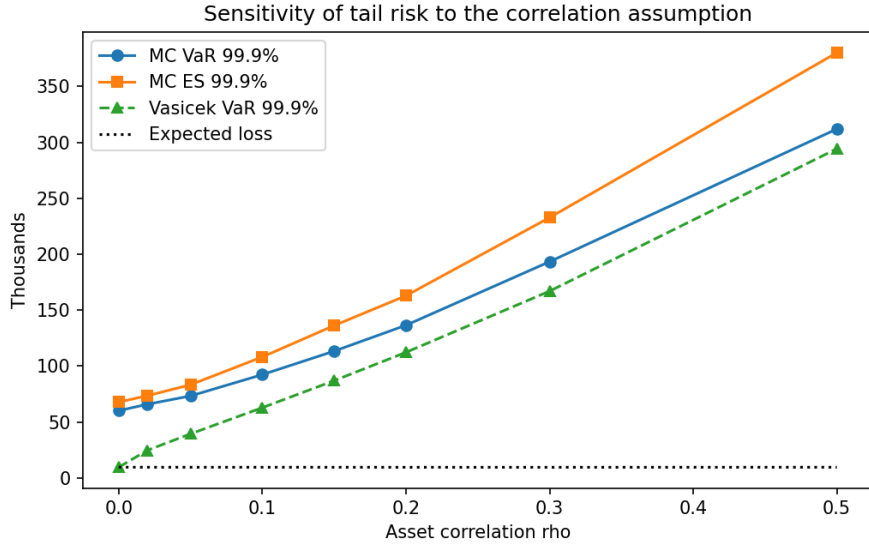


Figure 4: Simulated VaR and ES and analytical VaR at 99.9% as a function of the asset correlation ρ .

Scenario	EL	VaR 99%	VaR 99.9%	ES 99.9%
Baseline	9,562	70,991	124,726	150,099
PD $\times 2$	19,174	103,571	166,163	190,379
1-notch downgrade	17,220	99,497	161,010	183,979
LGD 60% (downturn)	12,749	94,654	166,302	200,132
PD $\times 2$ & LGD 60%	25,566	138,094	221,550	253,839

Table 8: Stress tests, 100,000 scenarios; Basel ρ recomputed from the stressed PDs. The baseline differs slightly from Table 6 because of the smaller number of scenarios.

The analytical Vasicek model assumes infinitely many small loans, so that idiosyncratic risk diversifies away completely. With only 60 loans it does not fully diversify, which creates a granularity gap of about 24,000 between the Vasicek estimate and the Monte Carlo result. In the extreme case $\rho = 0$ the analytical VaR collapses to the expected loss, while the simulation still shows a 99.9% VaR of 60,000 driven purely by idiosyncratic risk (Table 7). The Basel capital requirement and the simulated economic capital are close, at roughly 110,000 versus 114,000, because both measure unexpected loss in the same one-factor model; the remaining difference reflects the maturity adjustment and the fact that the Basel formula ignores granularity. In the stress tests (Table 8), doubling the probability of default or raising the loss given default to 60% each adds roughly 42,000 to the 99.9% VaR, and the combined scenario adds about 97,000. Since the VaR roughly doubles when the correlation moves from 0.10 to 0.30 (Figure 4), the correlation assumption is the single largest source of model risk.

6 Limitations and Future Work

The most significant limitation for this portfolio is correlation uncertainty: because the empirical data are uninformative, the results depend heavily on the assumed correlation. The Gaussian copula also implies zero tail dependence, which tends to underweight the joint extreme defaults observed during financial crises (McNeil et al., 2015). Keeping the loss given default fixed and independent of the systematic factor understates tail risk as well, since recovery rates typically fall in downturns. Finally, the PDs come from a generic, time-homogeneous transition matrix that is not specific to the healthcare sector. Future modelling should explore a t -copula to capture tail dependence, introduce a multi-factor model for healthcare sub-sectors, and link a stochastic LGD directly to the systematic risk factor.

7 Conclusion

The portfolio's expected loss is relatively small at around 9,600, but the required tail capital lies between 110,000 and 125,000 depending on the methodology, roughly 7.5% to 8.5% of the total exposure. The dependence between borrowers, rather than the individual default probabilities, is the primary driver of tail risk. I recommend holding about 114,000 in economic capital, in line with the simulated risk, actively monitoring the high-risk CCC- and B-rated names together with the largest exposures, and revisiting the correlation assumption when better data become available.

References

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